INCH-POUND

MIL-P-63313 (AR)
AMENDMENT 5
1 JULY 1990
SUPERSEDING
AMENDMENT 4
15 December 1988

MILITARY SPECIFICATION

PRIMER, ELECTRIC, M80A1
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-63313 (AR), dated 29 June 1979, and is approved for use within the U. S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete "(ALTERNATE)".

1.1: Delete "(Alternate)".

2.1: Specifications: MIL-E-46215: Change "Igniter, Electronic Primer, M63 and Electric Primers, M80A1 and M83" to "Electric Primers, M80A1, M83 and M120".

PAGE 2

2.1: Standards: Add "MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes".

Standards: Change: "MIL-STD-286 - Propellants: Sampling, Inspection and Testing" to "MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing".

Drawings: 8839499: Delete "Sheet 2-".

3.1.1: Delete "(Sheet 2)".

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* 3.3.1: Delete:

"...at the loading station at the time of loading the ignition element...",

and substitute: "...after drying...".

Delete: "...0.10 percent." and substitute: "...0.02 percent."

PAGE 4

4.2: Change "at" to "on".

4.3.1: Delete "(Sheet 2)".

PAGE 6

Table I: Examination or Test Block: Delete "-Sheet 2".

PAGE 8

* 4.4.2.1: Note 1: Delete in its entirety and substitute:

"Three samples minimum shall be selected from each drying oven after completion of drying cycle. If the moisture content of any sample exceeds the requirement, that quantity of material represented by the sample shall either be redried until acceptable results are achieved or rejected. The selected samples must not be dried more than any other unit which could be selected."

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4.4.2.3: Next Higher Assembly Block: Delete "(sheet 2)".

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* 4.4.2.4: Next Higher Assembly Block: Delete "(Sheet 2)".

Major 102: Delete: "100%" and substitute: "0.40%"

Delete: "105 Retainer Fails to Withstand Application of Minimum Assembly Torque. (See note if lacquer is used) 0.40% 3.2 4.5.8.1"

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4.4.2.5: Drawing Number Block: Delete "(sheet 2)".

Add Critical Defect:

"2	Plug missing	100%	3.2	Visual"

* Category 201: Delete: "Gage" and substitute "Visual".

PAGE 19

4.4.3.2.1: Change "Beginning with the first lot produced and continuing until three " to "Beginning with the first lot produced, or the lot following a rejected lot in 4.4.3.2.2, and continuing until three".

4.4.3.2.2: Add to end of paragraph: "If a lot is rejected under these criteria, continue testing lots in accordance with 4.4.3.2.1."

* 4.5.2.1: Delete in its entirety and substitute:

"4.5.2.1 Igniter charge composition. The moisture content of the igniter charge composition shall be determined as specified in Method 102.2.1 of MIL-STD-1234 except that the electric ignition element loading assembly and only a two hour drying time shall be used. The upper insulator from the electric ignition element loading assembly may be removed prior to analysis."

* 4.5.2.1.1: Delete in its entirety.

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4.5.2.3: Change "Method 102.1" to "Method 102.1.3".

4.5.3: Delete "- sheet 2".

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4.5.6, 4.5.6.1, 4.5.6.2, 4.5.6.3, and 4.5.6.4: Delete in their entirety and substitute the following:

"4.5.6 Determination of ignition charge composition. The chemical composition of the ignition charge shall be considered satisfactory if it is in accordance with the following:

Component	Percentage
Potassium Chlorate & Gum Solids	46.1 ± 2.1%
Lead Thiocyanate	35.5 ± 2%
Charcoal	20.0 ± 1.5%

4.5.6.1 Charcoal. Transfer approximately 2 grams of the igniter charge composition into a tared medium porosity crucible. Place in oven 70° C overnight to drive off water. Place crucible in desiccator to cool and then weigh to establish sample weight. Wash with ten 25 ml of distilled water. Use a small glass rod to break up any lumps in the sample during the extraction process. Preserve the filtrate for the determination for lead thiocyanate. Place in oven maintained at 100 ± 5° C for 2 hours, cool in a desiccator and weigh. Calculate the percentage of charcoal as follows:

$$\text{Percent charcoal} = \frac{(A-B)\ (100)}{A}$$

A = Beginning sample weight
B = Empty crucible weight

4.5.6.2 Lead Thiocyanate. Exactly 35 ml of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml of 3N nitric acid (nitric acid used should be water white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for five minutes, 5 ml of a saturated solution of ferric ammonium sulfate be added and the solution titrated with standardized 0.1N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

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$$\text{Percent lead thiocyanate} = \frac{16.17\ (AB-CD)}{E}$$

A = Volume of silver nitrate solution used
B = Normality of silver nitrate solution
C = Volume of potassium thiocyanate solution
D = Normality of potassium thiocyanate
E = Weight of sample

4.5.6.3 Potassium chlorate and gum solids. The percentage of potassium chlorate and gum solids is obtained by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent."

PAGE 22

5.1, 5.1.1, 5.2 and 5.2.1: delete in their entirety and substitute:

"5.1 Packaging and marking. Primers shall be packaged and marked in accordance with dwg. 8837899.

5.2 Packing and marking.

5.2.1 Level A. Packaged primers shall be packed and marked in accordance with dwg. 8837900.

5.2.2 Level B. Packaged primers shall be packed and marked in accordance with dwg. 8837900."

PAGE 23

6.1: Change "M392" to "M392A2".

Change "M724" to "M724A1".

6.5.1: Change "Method 102.1" to "Method 102.1.3".

6.5.2: Delete entirely and substitute:

"6.5.2 Electrical resistance. This test shall be performed in accordance with the applicable defect table. Any primer which fails to comply with the requirements specified in 3.4 shall be classed defective and removed from the lot. The test shall be performed as specified in 4.5.3."

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Figure 1: Title: Change "PAINT" to "LACQUER".

6.7: Change "M392" to "M392A2".

Change "M724" to "M724A1".

* 6.8: Change: "DRDAR-QAR-E and DRDAR-LCU-CV" to "AMSMC-QAT-A and SMCAR-LCE-D".

Change "01201" to "61299".

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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